

# 05 — Task Breakdown

|              |                                                                                                                                              |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| Version      | 0.1 (draft)                                                                                                                                  |
| Date         | 2026-09-27                                                                                                                                   |
| Based on     | BRD v0.6.1, PRD v0.6, HLD v0.2, LLD v0.1 (all approved); stack as proposed for Stage 6 (TypeScript, Next.js, Hono, Kysely, Supabase, Vercel) |
| Status       | AWAITING APPROVAL                                                                                                                            |
| Generated by | <code>tools/gen_tasks.py</code> (dependency check + critical path)                                                                           |

## 1. How to read this

- **Tracks:** a shared **Foundation** track first, then one track per service (each assignable to one developer — here, Sarang with Claude), then **Test & release**.
- **Size:** S  $\approx$  ½ day, M  $\approx$  2 days, L  $\approx$  4 days of focused work with Claude writing the code. Estimates, not commitments.
- **Parallel against mocks:** after the Foundation, every service track depends only on `libs/`, its own schema and **contract mocks** (F-05) — never on another service being built. Cross-service dependencies listed below are the few places where a real service replaces a mock (e.g. intake needs records' micromarkets).
- **Definition of Done** (CLAUDE.md §5) applies to every task; the last task of each service track is its explicit DoD check and ends with a stop for `APPROVED: <service>`.
- ★ = on the critical path.

## 2. Summary

| Track          | Tasks | Effort (days) |
|----------------|-------|---------------|
| Foundation     | 17    | 28.5          |
| records        | 12    | 32.5          |
| intake         | 11    | 23            |
| crm-engine     | 7     | 14.5          |
| journeys       | 11    | 27            |
| listings       | 7     | 12.5          |
| insight        | 7     | 16.5          |
| web            | 9     | 24            |
| Test & release | 6     | 12.5          |
| Total          | 87    | 191           |

**Critical path:** F-01 → F-02 → F-03 → F-04 → F-09 → JOU-01 → JOU-02 → JOU-03 → JOU-05 → JOU-07 → JOU-08 → JOU-11 → WEB-06 → WEB-07 → WEB-08 → WEB-09 → QA-01 → QA-02 → QA-03 → REL-01 → REL-02 → REL-03 = **55.5 working days**. This is the shortest possible elapsed time, reachable only if the service tracks run in parallel.

Elapsed time depends on how tasks run:

| Mode                                                                                   | Pilot ready (QA-02)  | Release (REL-03)       |
|----------------------------------------------------------------------------------------|----------------------|------------------------|
| One task at a time (CLAUDE.md Stage 7 as written)                                      | ≈ 184.5 working days | ≈ 191 working days     |
| Service tracks in parallel (Claude agents per service, you still approve each service) | ≈ 49–69 working days | ≈ 55.5–78 working days |

Sizes are deliberately conservative. Actual pace is measured after the Foundation track and this table is re-estimated then.

### 3. Stage 7 execution order

Service approval gates in this order (each ends with `APPROVED: <service>`): **records** → **intake** → **crm-engine** → **journeys** → **listings** → **insight** → **web**, then the pilot, then the paid-plan gate and `APPROVED: Release`. Records goes first because every other service consumes its events; web goes last because it integrates all cards (its shell and sign-in are built early inside its track once the Foundation is done).

Full task sequence (dependency-respecting): F-01, F-02, F-03, F-04, F-05, F-06, F-07, F-08, F-09, F-10, F-11, F-12, F-13, F-14, F-15, F-16, F-17, REC-01, REC-02, REC-03, REC-04, REC-05, REC-06, REC-07, REC-08, REC-09, REC-10, REC-11, REC-12, INT-01, INT-02, INT-03, INT-04, INT-05, INT-06, INT-07, INT-08, INT-09, INT-10, INT-11, ENG-01, ENG-02, ENG-03, ENG-04, ENG-05, ENG-06, ENG-07, JOU-01, JOU-02, JOU-03, JOU-04, JOU-05, JOU-06, JOU-07, JOU-08, JOU-09, JOU-10, JOU-11, LIS-01, LIS-02, LIS-03, LIS-04, LIS-05, LIS-06, LIS-07, INS-01, INS-02, INS-03, INS-04, INS-05, INS-06, INS-07, WEB-01, WEB-02, WEB-03, WEB-04, WEB-05, WEB-06, WEB-07, WEB-08, WEB-09, QA-01, QA-02, QA-03, REL-01, REL-02, REL-03

### 4. Tasks

#### Foundation

| ID   | Task                              | Acceptance criteria                                                                                                                                                                                                                                                                                                                                                                                                                                | Depends on | Size | CP |
|------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|----|
| F-01 | Monorepo scaffold                 | <code>services/&lt;svc&gt;</code> ×7 (incl. web), <code>libs/</code> , <code>contracts/</code> , <code>infra/</code> , <code>loadtests/</code> , <code>tools/</code> , <code>.github/workflows/</code> ; pnpm workspaces + Turborepo; CODEOWNERS maps every service folder to Sarang; each service has README, Dockerfile-equivalent build config, <code>src/{domain,application,adapters}</code> , <code>migrations/</code> , <code>tests/</code> | —          | S    | ★  |
| F-02 | Code quality baseline             | Shared TS config, ESLint, formatter; dependency-cruiser rules forbid domain → application/adapters and application → adapters imports (CLAUDE.md §3.1); gitleaks config; all pass on the empty scaffold                                                                                                                                                                                                                                            | F-01       | S    | ★  |
| F-03 | CI pipeline per service           | GitHub Actions per service with path filters: lint, typecheck, layer check, unit, integration (Supabase local), contract tests, secret scan, build; a failing check blocks merge                                                                                                                                                                                                                                                                   | F-02       | M    | ★  |
| F-04 | Contract tooling                  | OpenAPI lint + AsyncAPI validation in CI; TS types generated from <code>contracts/</code> ; <b>CI fails on drift</b> between code and contract (CLAUDE.md §3.3); <code>tools/gen_*</code> outputs checked in CI                                                                                                                                                                                                                                    | F-03       | M    | ★  |
| F-05 | Contract mocks for parallel work  | Mock server per OpenAPI spec (examples-based) + event fixture publisher from AsyncAPI, runnable locally; any service can run alone with its dependencies mocked (CLAUDE.md §3.9)                                                                                                                                                                                                                                                                   | F-04       | M    |    |
| F-06 | Local database stack              | Supabase CLI local stack; bootstrap SQL creates 7 schemas, <code>&lt;svc&gt;_owner/_migrator/_svc</code> roles, grants, pgmq/pg_cron/pg_net, PostgreSQL exposure off (data-hosting §2-3); pnpm dev starts everything                                                                                                                                                                                                                               | F-01       | M    |    |
| F-07 | Provision pilot + verify platform | Supabase Free project in <code>ap-south-1</code> and 7 Vercel Hobby projects in <code>bom1</code> via Terraform in <code>infra/</code> ; <b>verified</b> : pgmq, pg_cron, pg_net available; direct-connection limit recorded; Data API exposure none. Any failure → STOP and raise a CR (data-hosting §7)                                                                                                                                          | F-06       | M    |    |
| F-08 | libs/db                           | Pooled client through Supavisor (transaction mode, no prepared statements), per-role semaphore, tenant-scoped query helper, migration runner (forward-only), <code>idempotency_keys</code> helper (R-3); integration-tested on local stack                                                                                                                                                                                                         | F-06, F-02 | M    |    |

| ID   | Task                           | Acceptance criteria                                                                                                                                                                                                                                                                                | Depends on | Size | CP |
|------|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------|----|
| F-09 | libs/outbox and queue consumer | Outbox write in the same transaction; relay handler (SKIP LOCKED, fan-out to per-consumer pgmq queues from the AsyncAPI subscriptions); drain handler with <code>processed_events</code> dedupe, aggregateVersion ordering, backoff, DLQ after 5 attempts; tests prove at-least-once + idempotency | F-08, F-04 | L    | ★  |
| F-10 | libs/http                      | Hono app factory: correlation ID, RFC 7807 errors, input validation from generated schemas, Idempotency-Key middleware, If-Match, cursor pagination, health endpoints, outbound client (2 s timeout, retries with jitter on idempotent calls, circuit breaker)                                     | F-04, F-08 | M    |    |
| F-11 | libs/auth                      | Service-token verification against web JWKS (cached), staff header parsing, role + tenant re-check helpers, cron-secret and client-secret checks; unit-tested                                                                                                                                      | F-10       | S    |    |
| F-12 | libs/observability             | JSON logger with a PII allow-list (test proves phones/emails never logged), OpenTelemetry tracing across HTTP and events (transparent in envelope), RED metrics per route and consumer                                                                                                             | F-10       | M    |    |
| F-13 | libs/vocabulary                | Vocabulary release v0.6 files (BRD §4.2), runtime validators (R-11 trim/case-fold), generated display labels (For/Wants table), legacy-term translation table; exhaustive unit tests of the label table                                                                                            | F-02       | M    |    |
| F-14 | libs/redaction                 | Redaction of Indian phones, emails, names near contact phrases, unit/wing numbers; test set from anonymised real ad patterns; 0 leaks on the test set                                                                                                                                              | F-02       | M    |    |
| F-15 | Schedules and alarms           | <code>pg_cron</code> + <code>pg_net</code> job template (relay/drain every minute, service jobs) installed by migrations with per-service secrets; alarms: DLQ depth > 0, relay lag > 5 min, 5xx > 2%, p95 breach                                                                                  | F-09, F-07 | S    |    |
| F-16 | Backups and runbooks           | Nightly per-schema <code>pg_dump</code> to private Mumbai bucket (pilot) + monthly restore check; runbooks: deploy, rollback, rotate secrets, DLQ replay, restore                                                                                                                                  | F-07       | S    |    |
| F-17 | Synthetic data generator       | Vocabulary-driven generator of properties/offers/demands/people with fake contacts at pilot and production scale (5M) for tests and load tests; never uses real personal data                                                                                                                      | F-13       | M    |    |

## records

| ID     | Task                         | Acceptance criteria                                                                                                                                                                                                                                                                                            | Depends on               | Size | CP |
|--------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|------|----|
| REC-01 | Schema and migrations        | All tables/indexes of the records LLD §3 (incl. code sequences, source ads, sightings, merges + undo log, desks, market data, photos, vocabulary releases, micromarkets + adjacency, outbox, <code>processed_events</code> , <code>idempotency_keys</code> ); PII columns marked; tenant-first indexes         | F-08                     | L    |    |
| REC-02 | Reference data               | Vocabulary release load + <code>GET /v1/vocabulary*</code> + <code>vocabulary.released.v1</code> ; micromarkets CRUD with MMR seed + adjacency + <code>micromarkets.updated.v1</code> ; launch-area settings (R-9)                                                                                             | REC-01, F-13, F-09, F-10 | M    |    |
| REC-03 | Supply records API           | Properties, projects, offers CRUD/list/search with stored-field filters, cursor pagination, If-Match; <code>offer.*</code> , <code>project.*</code> events; codes issued per tenant                                                                                                                            | REC-02                   | L    |    |
| REC-04 | Demand, people and quick add | Demands, touches, people CRUD/search; quick-add lookup by phone hash (POST, R-10 stages); <code>demand.*</code> , <code>person.*</code> events; contacts masked by default                                                                                                                                     | REC-02                   | L    |    |
| REC-05 | Ingest from intake           | Consumer of <code>rows.classified.v1</code> : fetch PII via intake internal API (service token), apply migration map first, upsert by externalRef + content hash, source ads + split children, sightings, extractor-dedup trust, route non-Property scopes to desks, staff edits win, outside-launch-area flag | REC-03, REC-04, F-11     | L    |    |
| REC-06 | Dedup and merges             | Property-level supply dedup, demand touches (first touch credit), <code>possible_repeat_of</code> → merge candidates, price gaps; merge + column-level undo; <code>records.merged/merge_undone</code> , <code>merge_candidate.raised</code>                                                                    | REC-05                   | L    |    |
| REC-07 | Add supply and price sheets  | <code>POST /v1/demands/{id}/add-supply</code> (dedup first, Sourced-for tag, Contacted); project price sheets → <code>price_sheet.applied.v1</code>                                                                                                                                                            | REC-03, REC-04           | M    |    |
| REC-08 | Contact privacy              | <code>POST /v1/reveals</code> (audited, 60/h), <code>POST /internal/v1/contacts:batch</code> (R-21), <code>GET /internal/v1/properties/{id}/scan-terms</code> (R-20), <code>audit.recorded.v1</code>                                                                                                           | REC-04, F-11             | M    |    |
| REC-09 | Photos                       | Signed upload, attach/remove, internal signed-URL read; <code>photo.added/removed</code>                                                                                                                                                                                                                       | REC-03                   | M    |    |

| ID     | Task                        | Acceptance criteria                                                                                                                                                                                                                                                                                                                                                                                                                    | Depends on                                     | Size | CP |
|--------|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|------|----|
| REC-10 | Journey-driven consumers    | Consume <code>deal.closed</code> , <code>offer.retired</code> (market data + <code>market_data.recorded</code> ), <code>call.logged</code> (unreachable flag), <code>demand.exited/reactivated</code> , <code>lease_renewal.due</code> (Upcoming offer), <code>offer.confirmed</code> , <code>site_visit.completed</code> , <code>publication.changed</code> , <code>review_item.resolved</code> (reclassify / <code>*.voided</code> ) | REC-05, F-09                                   | M    |    |
| REC-11 | Desks                       | Desk lists + desk item patch; <code>desk_item.*</code> , <code>watchlist_item.created</code>                                                                                                                                                                                                                                                                                                                                           | REC-05                                         | S    |    |
| REC-12 | records: definition of done | Contract tests pass on all 69 operations; tenant-isolation test; README (run locally, env vars, owned data, events); metrics/traces/health; STATUS updated                                                                                                                                                                                                                                                                             | REC-06, REC-07, REC-08, REC-09, REC-10, REC-11 | M    |    |

## intake

| ID     | Task                           | Acceptance criteria                                                                                                                                                                                   | Depends on             | Size | CP |
|--------|--------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|----|
| INT-01 | Schema and migrations          | Tables/indexes of the intake LLD §3 (uploads, chunks, templates, <code>raw_rows</code> , <code>row_errors</code> , review items, outbox...)                                                           | F-08                   | M    |    |
| INT-02 | Upload API                     | Create (signed URL), get/list, progress, cancel, rejected-rows download; 5 uploads/h per user                                                                                                         | INT-01, F-10           | M    |    |
| INT-03 | Inspect, modes and templates   | Header inspection; strict mode only on exact 89-column match; mapping mode + templates CRUD; identical-file detection                                                                                 | INT-02, F-13           | M    |    |
| INT-04 | Split job                      | Streaming <code>xlsx/csv</code> → chunks (500 pilot / 2,000 paid), <code>migration_map</code> sheet parsed, chunk messages idempotent                                                                 | INT-03, F-09           | L    |    |
| INT-05 | Chunk worker: rules            | Strict validation (R-11, range-inverted), legacy translation, normalisation (units, localities via records micromarkets, BHK), reason codes (8), content hash + unchanged detection, raw rows written | INT-04, REC-02         | L    |    |
| INT-06 | Chunk worker: AI for leftovers | Redaction → Hugging Face call (batch 20) → vocabulary validation → <code>needs_review</code> on low confidence; fallback <code>model_unavailable</code> ; no PII leaves (test with intercepting mock) | INT-05, F-14           | M    |    |
| INT-07 | Pilot anonymiser               | Anonymise-on-import switch (CR-006 Z-9): consistent fake values, original file deleted after split                                                                                                    | INT-04                 | S    |    |
| INT-08 | Events and internal rows API   | <code>rows.classified</code> (≤500), <code>upload.*</code> , <code>review_item.created</code> ; GET <code>/internal/v1/uploads/{id}/rows?batch=</code> for records                                    | INT-05, F-11           | M    |    |
| INT-09 | Review queue                   | List/summary/resolve/bulk-resolve grouped by reason code; <code>review_item.resolved.v1</code>                                                                                                        | INT-08                 | M    |    |
| INT-10 | Parse for quick add            | POST <code>/v1/parse</code> (4 s), redaction before AI                                                                                                                                                | INT-06                 | S    |    |
| INT-11 | intake: definition of done     | Contract tests (30 ops); 20k-row pilot file processed ≤ 5 min locally; retention job (30 d pilot); README; metrics; STATUS                                                                            | INT-07, INT-09, INT-10 | M    |    |

## crm-engine

| ID     | Task                     | Acceptance criteria                                                                                                                                                                                              | Depends on   | Size | CP |
|--------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------|----|
| ENG-01 | Schema and projection    | Matchable PII-free projection of offers/demands from events (facts, lifecycle, commercial status, voids, merges, micromarkets, price sheets)                                                                     | F-08, F-09   | M    |    |
| ENG-02 | Matching core            | Hard filters and scoring (PRD §4.5), unknown basis ±25% + flag, unknown market, Stale reconfirm flag, date exclusions with reasons, outside-launch-area exclusion, top 20; unit tests from BRD §7 and Appendix B | ENG-01, F-13 | L    |    |
| ENG-03 | Bundles                  | 2–3 offers by buildingKey or adjacent micromarket (R-13), commercial/industrial only, combined area/price                                                                                                        | ENG-02       | M    |    |
| ENG-04 | Triggers and propagation | Incremental matching on events; price-change flags; close/reopen/superseded; <code>demand.matching_completed.v1</code> ; <code>match.*</code> events                                                             | ENG-02       | M    |    |
| ENG-05 | Match APIs               | Matches by demand/offer, detail + explanation, confirm/reject, exclusions, re-run, weights                                                                                                                       | ENG-04, F-10 | M    |    |

| ID     | Task                           | Acceptance criteria                                                                                 | Depends on             | Size | CP |
|--------|--------------------------------|-----------------------------------------------------------------------------------------------------|------------------------|------|----|
| ENG-06 | Jobs and feedback              | Nightly re-score sweep, projection reconcile, proposal feedback captured for M6 weight tuning       | ENG-04                 | S    |    |
| ENG-07 | crm-engine: definition of done | Contract tests (19 ops); matching precision checks on Appendix B scenarios; README; metrics; STATUS | ENG-03, ENG-05, ENG-06 | M    |    |

## journeys

| ID     | Task                                   | Acceptance criteria                                                                                                                                                                                                  | Depends on                                   | Size | CP |
|--------|----------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|------|----|
| JOU-01 | Schema and projections                 | Tables/indexes of the journeys LLD §3 incl. local projections from records/crm-engine/listings/web events                                                                                                            | F-08, F-09                                   | M    | ★  |
| JOU-02 | Life curve engine                      | Thresholds per category (BRD §4.5 + D-12 + R-12), <code>next_stage_at</code> , nightly job 02:00 IST, confirmation resets, stage actions, <code>lifecycle.stage_changed</code> , <code>offer/demand.confirmed</code> | JOU-01                                       | L    | ★  |
| JOU-03 | Queues                                 | Must/Should call ranking, demand queue sections, capacity (default 40), today list, reassign, <code>queue.counts_changed</code> flush ( $\leq 1/\text{min}/\text{user}$ ); My queue $p95 \leq 1$ s                   | JOU-02                                       | L    | ★  |
| JOU-04 | Calls and offer retirement             | Call outcomes, 3-attempt rule, <code>call.logged</code> , retire/reactivate (R-12), commercial axis for offers                                                                                                       | JOU-03                                       | M    |    |
| JOU-05 | Demand journey                         | Qualify, exits Lost/Dormant/Invalid (competing terms, person flag), reactivate, dormant revisits, commercial axis derivation                                                                                         | JOU-03                                       | M    | ★  |
| JOU-06 | Sourcing                               | Sourcing requests, anonymous demand post signal, needs-sourcing from <code>demand.matching_completed</code>                                                                                                          | JOU-05, ENG-04                               | M    |    |
| JOU-07 | Proposals                              | Snapshot from records, PDF (async) with MahaRERA number from listings, share link (14 d) + public page, mark sent, feedback (incl. "maybe")                                                                          | JOU-05                                       | L    | ★  |
| JOU-08 | Site visits and deals                  | Visits (reset both life curves), deals with mandatory next action/follow-up, close ( <code>deal.closed</code> , <code>unitsBooked</code> ), cancel compensation, lease renewal at month 10                           | JOU-07                                       | L    | ★  |
| JOU-09 | Work notifications and Watchlist tasks | Match/SRQ/handoff/follow-up notifications; Watchlist follow-up tasks for supply                                                                                                                                      | JOU-03                                       | S    |    |
| JOU-10 | Settings APIs                          | Life-curve thresholds, queue weights, capacities (Manager)                                                                                                                                                           | JOU-03                                       | S    |    |
| JOU-11 | journeys: definition of done           | Contract tests (60 ops); life-curve job on 5M synthetic subjects < 2 h; README; metrics; STATUS                                                                                                                      | JOU-04, JOU-06, JOU-08, JOU-09, JOU-10, F-17 | M    | ★  |

## listings

| ID     | Task                         | Acceptance criteria                                                                                                                                      | Depends on     | Size | CP |
|--------|------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|------|----|
| LIS-01 | Schema and projections       | Publication state, offers/projects/demand-post projections, change feed, API keys (hashed)                                                               | F-08, F-09     | M    |    |
| LIS-02 | Publication ceiling          | Ceiling computation, level set, auto-downgrade on Stale/Expired/Closed/Inactive/voided/merged, <code>publication.changed</code>                          | LIS-01, F-13   | M    |    |
| LIS-03 | Privacy scan                 | Blocking patterns (phones, emails, URLs, handles, units, exact floors, street) + building-name hashes via records scan-terms; photo text = warning (R-8) | LIS-02, REC-08 | M    |    |
| LIS-04 | Public projection            | Generated labels, PRD §8.3 fields (anonymous vs public), photo renditions via records signed URL into public bucket, MahaRERA + project RERA             | LIS-03, REC-09 | M    |    |
| LIS-05 | Public API                   | Listings/projects/demand posts/detail + change feed; cursor $\leq 50$ ; edge cache headers; per-key rate limits (R-1); withdrawal $\leq 1$ min           | LIS-04         | M    |    |
| LIS-06 | Admin                        | API keys create/rotate/revoke; publication settings (MahaRERA number), readable by journeys service token                                                | LIS-01, F-11   | S    |    |
| LIS-07 | listings: definition of done | Contract tests (25 ops); automated "0 PII in API output" scan (M8); README; metrics; STATUS                                                              | LIS-05, LIS-06 | M    |    |

## insight

| ID     | Task                        | Acceptance criteria                                                                                                                                                                               | Depends on             | Size | CP |
|--------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|----|
| INS-01 | Schema and read model       | Analytics tables fed by all subscribed events (63), conversations (redacted text only), plan catalogue, exports                                                                                   | F-08, F-09             | L    |    |
| INS-02 | Dashboards                  | Demand, supply, other scopes, data quality (classification grids, side checks, queue tiles from <code>queue.counts_changed</code> ); $\leq 2$ s p95                                               | INS-01, F-13           | M    |    |
| INS-03 | Query plans                 | Allowed query-plan catalogue + executor on the read model + "How I got this"                                                                                                                      | INS-01                 | M    |    |
| INS-04 | Chat                        | Redaction → Hugging Face plan → validation → template answers (no model-written facts), SSE streaming, keyword fallback, out-of-scope refusal, action cards referencing owning-service operations | INS-03, F-14           | L    |    |
| INS-05 | Exports                     | Async Excel (20k pilot / 100k prod), contacts via records batch for allowed roles (not Data operators), 24 h link, audit, <code>export.*</code>                                                   | INS-03, REC-08         | M    |    |
| INS-06 | Chat benchmark              | PRD Appendix A (13 questions) automated; M7 $\geq 85\%$ ; outbound AI requests checked PII-free                                                                                                   | INS-04                 | S    |    |
| INS-07 | insight: definition of done | Contract tests (20 ops); README; metrics; STATUS                                                                                                                                                  | INS-02, INS-05, INS-06 | M    |    |

## web

| ID     | Task                           | Acceptance criteria                                                                                                                               | Depends on             | Size | CP |
|--------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|------|----|
| WEB-01 | App shell                      | Next.js chat-first layout from the prototype: sidebar, home tiles, composer with "/" actions, side panel, card framework, light/dark              | F-01, F-02             | M    |    |
| WEB-02 | Sign-in and service tokens     | Supabase Auth Google (Workspace, invited only), session, <code>/v1/me</code> , service-token issuer + JWKS (R-2), client-secret auth for services | WEB-01, F-07, F-11     | M    |    |
| WEB-03 | Gateway                        | Routing table ( <code>x-routes</code> ) to services, auth + tenant check, token-bucket rate limits, correlation IDs, pass-through Idempotency-Key | WEB-02, F-10           | M    |    |
| WEB-04 | Users, audit and notifications | Users/invitations/roles + <code>user.changed.v1</code> ; audit sink + audit log; upload/export/user notifications merged with journeys            | WEB-03, F-09           | M    |    |
| WEB-05 | Cards: intake and records      | Upload (attach → mapping → progress → report), review queue, quick add (classification order), add supply, record panels, reveal, desks           | WEB-03, INT-11, REC-12 | L    |    |
| WEB-06 | Cards: engine and journeys     | My queue panel, call outcome, qualify, matches + bundles, sourcing, proposal, site visit, deal, exit, close/retire                                | WEB-05, ENG-07, JOU-11 | L    | ★  |
| WEB-07 | Cards: listings and insight    | Publication card, chat with streaming + action cards + "How I got this", dashboards panel, exports                                                | WEB-06, LIS-07, INS-07 | L    | ★  |
| WEB-08 | Settings page                  | Users, capacities, thresholds, weights, micromarkets, vocabulary (read-only), MahaRERA, API keys, audit log                                       | WEB-07                 | M    | ★  |
| WEB-09 | web: definition of done        | Playwright E2E for the core cards; accessibility checks (WCAG 2.1 AA core flows); README; metrics; STATUS                                         | WEB-08                 | M    | ★  |

## Test & release

| ID     | Task                            | Acceptance criteria                                                                                                                                                   | Depends on        | Size | CP |
|--------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|------|----|
| QA-01  | End-to-end acceptance scenarios | PRD Appendix B AS-D1...D4 and AS-S1...S6 pass across all services on the local stack                                                                                  | WEB-09            | L    | ★  |
| QA-02  | Pilot deployment                | All 7 deploy to the pilot; anonymised import of the real extractor master (2,155 rows); smoke tests; demo to 11 Estates                                               | QA-01, F-15, F-16 | M    | ★  |
| QA-03  | Pilot benchmarks                | M3 dedup precision, M4 repost recall, M5 categorisation, M6 match relevance collected on pilot data; pilot load smoke (capacity plan §8)                              | QA-02             | M    | ★  |
| REL-01 | Paid-plan gate                  | Vercel Pro + Supabase Pro + PITR (Mumbai) + dedicated Hugging Face endpoint provisioned; standby/failover checked vs NFR-3 (CR if needed); real data enabled only now | QA-03             | M    | ★  |

| ID     | Task                 | Acceptance criteria                                                                                                              | Depends on   | Size | CP |
|--------|----------------------|----------------------------------------------------------------------------------------------------------------------------------|--------------|------|----|
| REL-02 | Production load test | k6 scenarios L1–L6 (1,000 rps listings, staff mix, 100k upload ≤ 30 min, 5M life curve, chat, 2 h soak) pass on production sizes | REL-01, F-17 | M    | ★  |
| REL-03 | Release              | Alarms fired in a drill, restore drill ≤ 1 h, runbooks complete; results reported → APPROVED: Release                            | REL-02       | S    | ★  |

## 5. Risks to the plan

| Risk                                                          | Effect               | Mitigation                                                                    |
|---------------------------------------------------------------|----------------------|-------------------------------------------------------------------------------|
| Free-plan limits (pgmq/pg_cron missing, function time limits) | Foundation blocked   | F-07 verifies first; stop and raise a CR on failure                           |
| Hugging Face free credits too small for testing               | AI tasks slowed      | Rules first; intercepting mock for tests; small paid top-up is a PO decision  |
| Dedup thresholds need real samples                            | M3/M4 targets missed | Tunable thresholds; QA-03 measures on the real (anonymised) master            |
| Contract changes discovered during build                      | Rework               | Change Request per CLAUDE.md; contracts regenerated by tools; drift fails CI  |
| Single builder                                                | Long elapsed time    | Strict order; mocks let Claude work on one service without waiting for others |

## 6. Assumptions

- The Stage 6 stack is as proposed; if Stage 6 changes it, tasks keep their scope and only tooling words change.
- Vinit keeps producing extractor files in the approved 89-column schema (PRD Appendix C).
- A sample WhatsApp extractor file arrives before INT-11 (PRD OQ-P11) to confirm sender fields.